

UME404 : Mechanics of Deformable Bodies

L	T	P	Cr
3	1	0	3.5

Course Objective: The objective for this course is to develop the basic skills and knowledge required to analyze displacement field, stress, strain and failure in deformable solids using analytical solutions. This course also introduces an overview of important structural engineering design philosophies. This understanding will include concepts such as curved beam, unsymmetrical bending, helical spring, pressure vessel etc.

Syllabus

Three-Dimensional Stress Analysis: Stresses on an arbitrary plane, principal stresses and stress invariant, Mohr Circle in 3D, differential equations of equilibrium in Cartesian and cylindrical coordinates.

Three-dimensional strain analysis: rectangular strain components, (*Flipped learning component: principal strains and strain invariant*), compatibility conditions.

Stress-Strain Relations: Generalized Hooke's law, stress-strain relations for isotropic materials.

Theories of Elastic Failure: Various theories of failure, significance and applications.

Energy Methods: Principle of superposition, work done by forces- elastic strain energy stored, Maxwell-Betti's theorem, Castigliano's theorems, strain energy expressions, fictitious load method, statically indeterminate problems.

Brief introduction of Unsymmetrical Bending, Bending of Curved Flexural Members, Stress distribution in Thick Cylinders and Rotating Discs (*Flipped learning component: Thin cylinder and Helical Springs*).

Introduction to Plasticity: Stress-strain curve and constitutive relations for plastic flow, Prandtl–Reuss equations, Saint Venant–von Mises theory of plasticity.

Assignment: Students in groups will work on the problem formulation and perform failure analysis of different mechanical and structural components, such as model beams, pressure vessel, explosion of a pipeline, automobile suspension, crane hook, tail pipe support etc. Modelling of these problems on any available finite element software and comparison of the results with analytical solutions. This assignment also includes technical report writing and viva voce.

Course Learning Objectives (CLO)

The students will be able to:

1. Calculate the state of stress at the critical point of the object.
2. Establish 3D stress-strain relationship for isotropic materials.
3. Conduct the failure analysis under static loading in ductile and brittle materials using different theories of failures.
4. Calculate deflection at any point on a solid structure using Castigliano's theorems.

5. Determine the distribution of circumferential and radial stress along the thickness of thick cylinders.
6. Model and analyze real structures or engineering systems through projects/assignments

Text Books

1. Srinath, L.S., Advanced Mechanics of Solids, Tata Mc-Graw Hill (2005).
2. Popov, E.P., Engineering Mechanics of Solids, Prentice Hall of India (2006).
3. Boresi, A.P. and Schmidt, R.J. Advanced Mechanics of Materials, Wiley (2002).

Reference Books

1. Shames, I.H., Mechanics of Deformable Solids, Prentice Hall of India (2000).
2. Kumar K. and Ghai, R. C., Advanced Mechanics of Materials, Khanna Publishers (1986).
3. Ryder, G.H., Strength of Materials, B.I. Publishers (2005).

Evaluation Scheme

S. No.	Evaluation Elements	Weightage (%)
1.	MST	30
2.	EST	45
3.	Sessionals (Project/Assignment/Tutorials/Quizzes/Flipped learning component)	25